



Project Report

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1 Executive Summary

Our two-phase linear optimization project seeks to improve bike allocation and rebalancing operations at BIXI, Montreal's native bike-sharing system. BIXI's success relies on ensuring that riders consistently find available bikes for rental and open docks for returns. As demand fluctuates by time and location in the city, the system can often become unbalanced over the course of a day, leading to shortages, lost trips, and diminished user experience. Our goal was to leverage linear optimization to minimize these shortages and identify the most efficient rebalancing route for BIXI's truck fleet.

Phase 1 establishes the optimal baseline bike distribution across the city. After cleaning and filtering of the 2024 BIXI trip dataset, stations were grouped into weighted geographic clusters to create stable demand profiles. Using outward trip demand during peak periods, we formulated a linear optimization model that allocates the fixed fleet of 10,000 bicycles across weekday and weekend conditions. The model minimizes network shortages while enforcing fairness constraints to prevent chronic undersupply in any cluster. The resulting allocation provides a benchmark inventory level for each cluster at the start of a typical day.

Phase 2 determines the shortest route for a rebalancing truck to restore these optimal cluster inventories after daily usage. Using coordinates for each cluster, Manhattan distances were computed between cluster nodes, with decision variables representing truck routing, bike pickups and drop-offs, and load flow. A Miller–Tucker–Zemlin formulation was leveraged to prevent subtours to ensure a single continuous route. Using demand changes observed on a representative weekday, the model identifies the minimal distance traveled that repositions surplus bikes into shortage areas. The resulting travel route provides a realistic blueprint for nightly inter-cluster redistribution before intra-cluster balancing occurs.

Together, these two phases help solve a critical problem for BIXI: a precise target distribution of bicycles and a minimal route distance calibration plan to maintain it. Our approach is grounded in real user demand patterns and directly supports more efficient use of resources.

2 Introduction

BIXI is a public bicycle sharing system that has become an integral part of the urban community in the greater Montreal area. First launched in 2009, it holds the distinction of being North America's first large-scale bike sharing program, setting a precedent for similar initiatives adopted in major cities around the world, including London, New York and Chicago. With its innovative technology and modular design, BIXI was among the first to integrate advanced features like solar-powered docking stations and RFID tags for tracking, which helped define the modern, scalable bike-share model that is now adopted globally. In fact, just a month ago, BIXI was listed in TIME Magazine's Best Inventions Hall of Fame of the top 25 inventions in 25 years, placing the Montreal native bike sharing program alongside the iPhone as innovations that have had a remarkable impact on humanity.

The system is designed to allow users to rent a bike from one station and return it to any other location across the network, offering a convenient, on-demand, and eco-friendly transportation alternative for short urban trips. Being a key pillar of Montreal's urban mobility strategy, BIXI handles millions of trips annually and is increasingly seen as a true mode of public transport. However, this high volume of one-way travel creates a constant operational challenge: system imbalance due to the supply and demand of bikes at any given station.

Effectively managing the redistribution of these bicycles is critical to the system's success, to ensure that a bike is always available where and when a rider needs it. The operational reality is that an unbalanced system directly translates into lost revenue and customer dissatisfaction. If a potential rider finds no bikes available at a station or cannot return a bike due to no available docks, a rental trip is lost, and the user experience is degraded. Thus, effective reallocation of bikes between stations helps minimize these potential service failures and encourages higher subscription rates and more repeat usage. Today, the primary method for station calibration involves a dedicated fleet of trucks travelling around the city to rebalance stations when required. Alternatively, BIXI encourages users to help rebalance the network by taking a bike from a full station and/or returning a bike to an empty station in exchange for rewards and points through its AMI BIXI program.

3 Problem Statement

Our foundational linear optimization model seeks to minimize total shortage in the network, across all days, clusters, and peak periods. Phase 2 of our problem seeks to find the most efficient route and rebalancing plan that minimizes total travel distance. There were several compelling reasons why the problem of optimizing BIXI's bike allocation was specifically chosen for our linear optimization project.

First, the reallocation challenge maps perfectly into a network flow model, where the objective is to move the necessary flow from oversupplied nodes to undersupplied nodes. A rigorous mathematical approach is needed to guarantee the minimum shortage solution, which is precisely what linear optimization through tools like Gurobi can offer.

Second, linear optimisation requires constraints to be linear, which our reallocation problem satisfies. Constraints like “sum of bikes across clusters must equal to total bikes in the system”, or “shortage equals demand minus assigned bikes” can be easily expressed as linear inequalities.

Third, linear optimization requires a single, measurable objective, which is to minimize or maximize a function. Across the two distinct phases of our problem statement, the goal is clearly defined as minimizing total shortage across the network and minimizing total travel distance between nodes during reallocation.

4 Data Pre-Processing

4.1 Data Sanitation and Temporal Filtering

The primary dataset for this analysis was obtained from the BIXI Montreal Open Data portal, utilizing the available trip logs for the 2024 season. To ensure the operational model relied on valid patterns, the raw data was subjected to a cleaning pipeline to standardize metrics and remove anomalies.

First, column names were standardized to lowercase and stripped of whitespace. Timestamps were converted to datetime objects, and a calculated "hour" column was created with a -5 hour offset to align with Montreal Local Time (EST/EDT). This standardization allowed for the creation of

specific boolean flags for "rush hour" windows (AM: 06:00-10:00 and PM: 16:00-20:00) and a categorical flag for weekends (is_weekend).

We then applied a duration-based filter to the trip data. Trips with a duration of less than 1 minute were removed to filter out false starts or mechanical checks. Conversely, trips exceeding 180 minutes were excluded as outliers, as they do not reflect the standard point-to-point transit behavior this study aims to optimize. Additionally, timestamp validity checks were enforced to ensure all records fell within a realistic year range (2020–2030).

For the clustering and optimization phases, the analysis focused on a specific target month to ensure consistency in demand modeling. Based on the configuration in our code, November (Month 11) was selected as the TARGET_MONTH. The dataset was filtered to include only trips occurring in this specific month. This filtration provided the distinct dataset (df_month) used for the subsequent clustering and linear optimization steps.

4.2 Station Featuring

Optimizing rebalancing routes for the full network of over 1200 stations presents a computationally intractable problem. To address this, we aggregated raw transactional data into a station-level feature matrix to facilitate clustering. Data was grouped by ‘start station name’, and for every station, the following metrics were calculated.

1. Demand Intensity: The total count of outbound trips (starts). This metric serves as the weighting factor in the clustering phase.
2. Net Flow: Calculated as Ends – Starts, this identifies if a station generally accumulates bikes or loses them.
3. Temporal Profile: Boolean flags for AM and PM peaks were summed to quantify the station's role in the daily commute
4. Average Duration: The mean duration of trips originating from ‘s’ was computed to distinguish between leisure-oriented stations and transit-oriented stations.

Finally, a geospatial bounding box was enforced to handle data errors. Stations with missing coordinates or those falling outside the range of Latitude 40°–50° and Longitude -80° to -60° were dropped from the feature matrix to ensure all analysed nodes fell within the valid Greater Montreal service area.

4.3 Weighted Spatial Clustering and Node Identification

The segmentation of the BIXI network into discrete operational zones was achieved using a Weighted K-Means algorithm with the number of clusters (`N_CLUSTERS`) set to 30. We chose this number through iterative testing to identify the most effective model configuration. It provided enough detail to accurately capture neighbourhood demand patterns, but kept the problem simple enough for our optimization model to solve without issues.

Unlike standard K-Means, this approach assigned a sample weight to each station corresponding to its total trip starts (`weights = station_features["starts"]`). This weighting ensures that the resulting centroids are "gravitationally" pulled toward high-demand stations rather than being located in the geometric center of a sparsely populated area.

For each of the 30 defined clusters, a "Cluster Node" was identified by calculating the mean latitude and longitude of its constituent stations. These centroids serve as the representative nodes for the "Hub-and-Spoke" logistics planning in Phase 2.

5 Baseline Bike Allocation (Phase 1)

Phase 1 builds the starting point for the rest of the project: how many bikes we would ideally like to have in each part of the city at the beginning of a typical day. Concretely, this phase does two things. First, it translates raw trip records into a demand base at the cluster level. Second, it uses a linear optimization model to allocate a fixed fleet of bikes across those clusters so that expected shortages are as low as possible on both weekdays and weekends.

5.1 Constructing a demand base at cluster level

Once every station has a `geo_cluster`, we attach this label back to the trip-level data via the start station. We then focus on outward demand only (trips starting in a cluster), because these are the trips that actually require bikes to be sitting in docks at the start of a period. For each calendar date `t` in the month, each cluster `c` and each peak period `p` (AM and PM), we count how many trips start there. These counts are stored in a three-dimensional array `demand_3d_month[t, c, p]`. Intuitively, `demand_3d_month[t, c, p]` is our estimate of how many bikes riders would like to take out of cluster `c` in period `p` on day `t`.

Finally, we classify each calendar day `t` as either a weekday or weekend day and record its day type in `day_type_of_t[t]`. This allows the model to distinguish between two behaviour patterns (`D = {0 = weekday, 1 = weekend}`) while still using the same monthly demand base.

5.2 Phase 1 Optimization Model

Phase 1 builds the “starting point” for the entire project: how many bikes we would ideally like to have in each part of the city at the beginning of a typical day, before any reallocation by truck. This phase has two main steps. First, we simplify the BIXI network into a fixed number of geographic clusters. Second, we use historical trip data to estimate outward demand from each cluster and then build a linear optimization model that allocates a fixed fleet of bikes across those clusters while limiting total shortage.

5.3 Decision Variables and Objective Function

Our Phase 1 model contains the following decision variables:

- $B[d, c]$: An integer variable representing the number of bikes assigned to cluster c on a day of type d , where $d=0$ is weekday and $d=1$ is weekend. This defines the baseline number of bikes available in each cluster for that day type.
- $s[t, c, p]$: A continuous variable (with lower bound 0) representing the shortage in cluster c , on day t , during period p , where $p=0$ is AM peak and $p=1$ is PM peak

The key parameters driving the model are:

- $\text{demand_3d_month}[t, c, p]$: outward demand from cluster c on day t in period p ,
- $\text{TOTAL_BIKES} = 10\,000$: the fixed fleet size,
- $\text{day_type_of_t}[t]$: the mapping from each day to its day type, and
- $\text{cluster_total_demand}[c]$: total outward demand over the month in cluster c .

The model’s objective is to minimize the total shortage across all days, clusters and periods, i.e., the sum of $s[t, c, p]$ over all t, c and p . In other words, every unserved trip counts equally in the objective, regardless of where or when it occurs.

5.4 Constraints

Our Phase 1 model contains three general categories of constraints.

1. **Bike fleet constraints:** For each day type d (weekday and weekend), the total number of bikes assigned across all clusters must equal the fixed fleet size TOTAL_BIKES .
 - Formally, for each d we impose $\sum(c) B[d, c] = \text{TOTAL_BIKES}$. This ensures that the model simply reallocates a fixed pool of bikes rather than creating or destroying them.

2. **Shortage definition constraints:** For every calendar day t , cluster c and peak period p , shortage must cover any gap between demand and the bikes assigned to that cluster for the corresponding day type. Let $d = \text{day_type_of_t}[t]$. We impose $s[t, c, p] \geq \text{demand_3d_month}[t, c, p] - B[d, c]$.
 - If $B[d, c]$ is large enough to cover $\text{demand_3d_month}[t, c, p]$, the constraint allows $s[t, c, p]$ to be zero. If $B[d, c]$ is too small, the difference appears as positive shortage. This is how the high-level allocation $B[d, c]$ is connected to the day-by-day demand patterns.
3. **Fairness constraints at cluster level:** A pure shortage-minimization model could, in theory, choose to “sacrifice” a few clusters entirely if this improves performance elsewhere. To prevent this, we introduce a fairness cap with parameter $\text{TAU_CLUSTER} = 0.20$. For each cluster c with positive total demand, we restrict its aggregate shortage across the month to be at most 20% of its total demand:
 - $\sum_{(t,p)} s[t, c, p] \leq \text{TAU_CLUSTER} \times \text{cluster_total_demand}[c]$

5.5 Optimization Results

After solving the model in Gurobi, we obtain $B[d, c]$ for both day types. By weighting these by the number of weekdays and weekends in the month, we also compute an average “target” number of bikes per cluster. This Phase 1 baseline allocation serves as the starting distribution for the system and provides the benchmark that Phase 2’s truck routing model attempts to restore at the end of each operating day. See Appendix A.

6 Route Optimization (Phase 2)

Phase 2 involved optimizing the route taken by a truck to redistribute bikes between clusters to meet their optimal allocations. We envisaged a two-part operation for the reallocation. First, bikes are reallocated between clusters (inter-cluster or meta-reallocation) to reverse changes to the optimal bike allocation as result of trips by made users. This is what we modelled. This process would begin after the peak PM demand period, around 8 pm. After the inter-cluster reallocation, the bikes can be further reallocated between the docking stations within the cluster (intra-cluster reallocation; not modelled). Inter-cluster reallocation is a necessary precursor to intra-cluster allocation, because cluster with a shortage will not have sufficient bikes for distribution to stations.

6.1 Locate cluster coordinates, calculate inter-cluster differences and bike reallocations

First, we to spatially located each cluster. For simplicity, we set the location as the average coordinates of the cluster's underlying stations. A map showing the approximate location of every cluster is in Appendix B.

Once each cluster had coordinates, we calculated the “Manhattan distance” between each cluster. This is the sum of the absolute differences between the latitudes and longitudes of each cluster-pair, calculated by moving only along axes at right angles. It assumes that Montreal streets are set up as a perfect grid, which is not the case. In reality, the shortest possible route that a truck can travel between two points will be a bit longer than the Manhattan distance, but it is nevertheless much more accurate than using straight-line distance between both cluster nodes (i.e., “as the crow flies”). This information was stored in a matrix, with columns and rows indexed by cluster ID, allowing the distance between any cluster pair to be retrieved with the “ $x[i][j]$ ” approach.

For each cluster, we calculated the difference in the starting trips and ending trips between 6 am and 8pm on a typical weekday (Wednesday, November 13, 2024). These times were chosen because most trips happen within that window, and now the City has a window of low activity to perform the reallocation. We do not have information on the number of bikes within the cluster at any moment, but we assume that the number of bikes at 6 am should resemble our optimal bike distribution from Phase 1. These differences (“net_change”) were incorporated into a new dataframe called “bike_flow”, which informs the model's reallocation constraints.

If more bikes arrive than leave throughout the day, that cluster has a surplus. If more bikes leave than arrive, that cluster has a shortage. We incorporated a 5% margin around the net_change, which provided the upper and lower limits for pickups and drop-offs at each cluster. The rationale for this margin is to provide the model with some flexibility, albeit narrow, to find the shortest route.

6.2 Decision Variables and Objective Function

Our model contains the following decision variables:

- $x[i, j]$: A binary variable that is 1 if the truck travels from cluster i to cluster j , and 0 otherwise. This defines the trips that the truck must take. It is subject to several constraints.
- $p[i]$: A continuous variable representing the number of bikes picked up at cluster i .

d[i]: A continuous variable representing the number of bikes dropped off at cluster i.

f[i, j]: A continuous variable representing the number of bikes carried by the truck when travelling from cluster i to cluster j.

u[i]: A continuous variable used for subtour elimination (see details under Constraints).

Although p, d and f should notionally be integer variables since bikes exist in discrete units, we nevertheless chose to define them as continuous to lighten the computational load on the model and allow more flexibility to find a solution. The difference between, say, 100.5 bikes and 100 bikes is negligible in the overall solution quality. After the optimization, these continuous values can be rounded to the nearest integer. The model's **objective** is to minimize the total distance travelled by the truck, calculated as the sum of the distance between each cluster-pair where $x[i, j] = 1$.

6.3 Constraints

Our model contains three general categories of constraints.

- 1) **Bike reallocation constraints**: ensure that the number of bikes picked up or dropped off at each cluster stays within the lower and upper limits defined in the bike_flow dataframe.
- 2) **Truck-related constraints**:
 - a. Truck flow: Logically, the number of bikes in the truck at any point in the route must be connected to the number of bikes that it has picked up or dropped off. For each cluster i, the bikes carried *into* i ($f[j, i]$), plus any bikes picked up ($p[i]$), minus any bikes dropped off ($d[i]$), must equal the bikes carried *out of* i ($f[i, j]$).
 - b. Truck capacity: for any segment that is part of the route ($x[i, j]$), the number of bikes inside the truck ($f[i, j]$) does not exceed the truck capacity (set to 1000 bikes).
- 3) **Routing constraints**:
 - a. Continuity: the truck enters and leaves each cluster exactly once, forming a continuous tour. By setting each of the sum of $x[j, i]$ and of $x[i, j]$ as equal to 1 for every cluster-pair where i is not j, we ensure that the truck will take exactly one road segment leading into and out of each cluster.
 - b. Subtour elimination: Under the continuity constraint above, we could have disconnected continuous “subtours” (e.g., $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ and $4 \rightarrow 5 \rightarrow 6 \rightarrow 4$). However, this is impossible, since the truck cannot simply appear at a new sub-route, it needs to

travel there. To prevent subtours, we used a Miller-Tucker-Zemlin (MTZ) Formulation. In this formulation, $u[i]$ is a “position” variable indicating when cluster i is visited on the tour sequence. The constraint $u[i] - u[j] + \text{big_M} * x[i, j] \leq \text{big_M} - 1$ links the visit order u with the routing decision x . If the truck travels from i to j (so $x[i, j] = 1$), the constraint simplifies to $u[i] - u[j] \leq -1$, which implies $u[j] \geq u[i] + 1$, meaning that j must be visited *after* i in the tour. If $x[i, j] = 0$, the constraint becomes $u[i] - u[j] \leq \text{big_M} - 1$, which is loose and doesn’t impose any real constraints. Cluster 0 was used as the starting point and thus excluded from this constraint.

6.4 Optimization Results

The minimum distance required to perform the reallocation was 114.37 km. To see the ordered sequence of trips, first we compiled a list of all trips ($x[i, j] = 1$). We set a “current” variable as 0 (initial value of i). Then we looped through the trips list until we found the trip where i equals the current variable, append i and j to the sequenced list, set “current” as j , and repeated until the sequence was complete. The complete truck tour sequence was: 0→14 | 14→26 | 26→1 | 1→29 | 29→21 | 21→8 | 8→4 | 4→12 | 12→28 | 28→5 | 5→11 | 11→25 | 25→2 | 2→16 | 16→22 | 22→24 | 24→3 | 3→9 | 9→15 | 15→17 | 17→10 | 10→19 | 19→7 | 7→6 | 6→27 | 27→23 | 23→18 | 18→20 | 20→13 | 13→0.

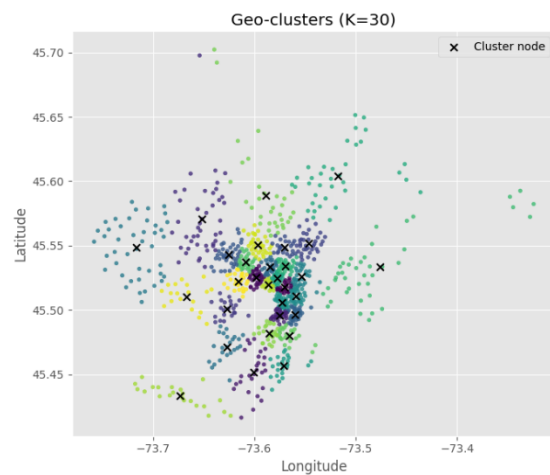
7 Appendix

Appendix A – Phase 1 Optimization Results

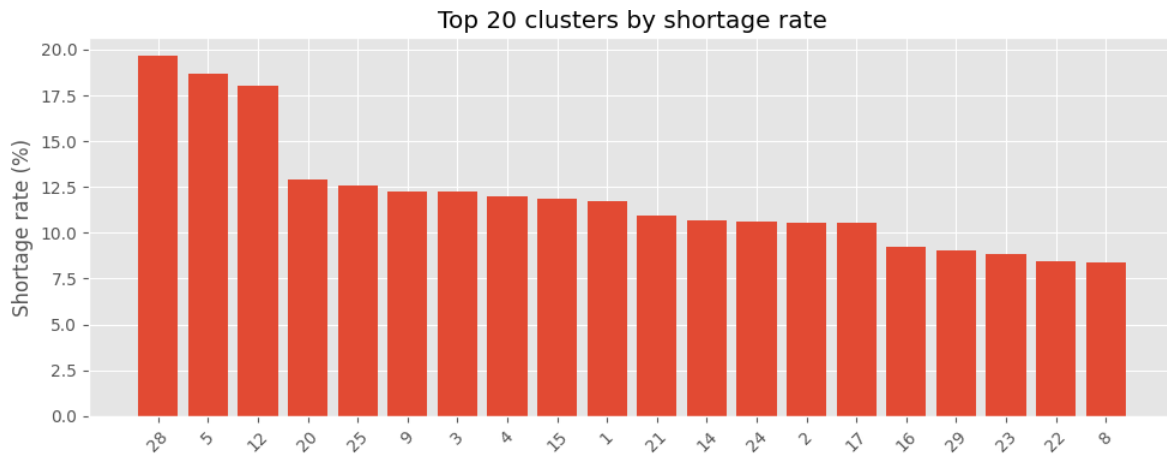
geo_cluster	bikes_weekday	bikes_weekend
0	580	543
1	584	612
2	163	139
3	652	474
4	145	121
5	241	174
6	426	384
7	277	364
8	309	394
9	507	457
10	472	579
11	138	156
12	33	24
13	382	399
14	537	415

geo_cluster	bikes_weekday	bikes_weekend
15	936	686
16	155	188
17	618	685
18	51	55
19	609	720
20	68	88
21	412	544
22	221	248
23	100	96
24	298	288
25	23	26
26	494	549
27	276	338
28	60	28
29	233	226

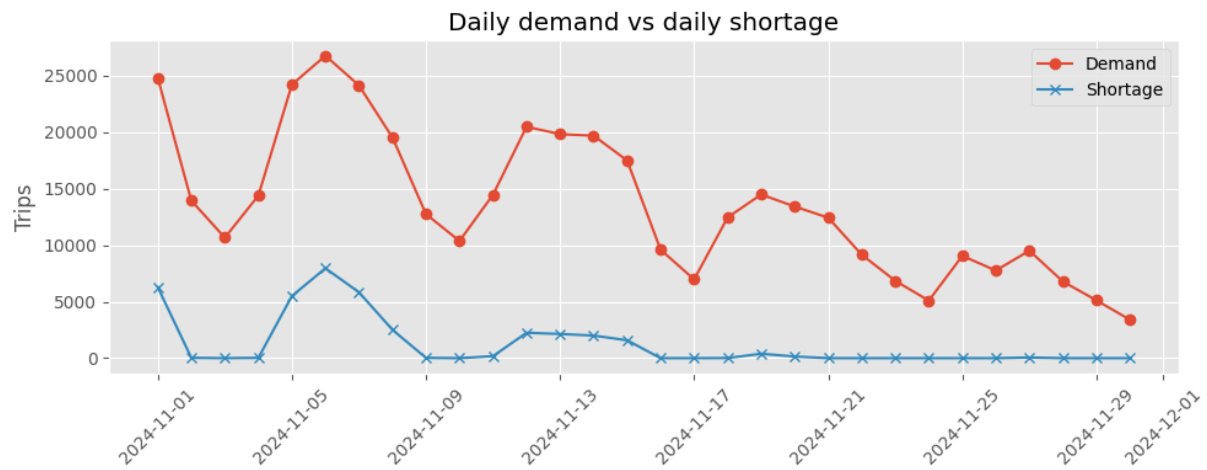
Appendix B – Cluster Location Map



Appendix C – Top 20 clusters by shortage rate



Appendix D – Demand Vs Shortage



Appendix E –End of day truck re-allocation plan

geo_cluster	pickup_plan	dropoff_plan
0	0	201
1	0	220
2	0	12
3	0	87
4	24	0
5	0	99
6	142	0
7	0	119
8	0	39
9	556	0
10	0	278
11	0	28
12	1	0
13	189	0
14	900	0

geo_cluster	pickup_plan	dropoff_plan
15	287	0
16	21	0
17	0	51
18	14	0
19	0	395
20	9	0
21	0	97
22	61	0
23	0	32
24	0	7
25	1	0
26	0	419
27	0	96
28	12	0
29	0	37

Appendix F – Top 15 clusters Before Vs After (Truck re-allocation)

